Don Fox
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Summary:

Thirty years of professional software development spanning telecommunications, warehousing, healthcare IT, and data systems. Primary focus has been backend engineering on Unix/Linux platforms, with experience across a broad range of languages, databases, and system architectures. More recent work includes web development, functional programming, and applied AI/prompt engineering. Proof-of-concept development — particularly in healthcare IT — has been a recurring part of the work, as has occasional intern mentoring.

I've mentored interns and worked on collaborative projects, over many occasions. Currently exploring AI engineering and functional programming with Elixir and Haskell. Available for remote or freelance opportunities.

Education:

- **Bachelor of Arts in Philosophy, 1982**
  University of Tennessee, Knoxville TN

Technical Training

- **Associate of Science Degree in Computer Science Technology, *Cum Laude,* 1991**
  Pellissippi State Technical Community College, Knoxville TN

Technical Skills

Languages: Python, C, C++, Perl, Elixir, Haskell, PL/SQL, C#
Frameworks: Django, Flask, Phoenix, SQLAlchemy,
Databases: PostgreSQL, MySQL, Oracle, MongoDB, MariaDB, Redis, Informix, DB2
Platforms: Linux, Unix, macOS, Solaris, AIX, Windows
Tools & Technologies: Git, Docker, Node.js, Shell Scripting, RESTful APIs.

Professional History:

EMRTS
https://emrts.us/

Software engineer working with a healthcare IT startup across six engagements, developing open-source tools for medical records, billing, provider enrollment, and claims. Work has centered on proof-of-concept development and R&D, often in collaboration with academic and professional partners. Contracts listed below fill the gaps between EMRTS engagements.

- **July 2025-Present:** Developing a proof-of-concept application for a Provider Vault medical enrollment website. Combines Elixir/Phoenix for scalability and concurrency with Python components for AI prompting to explore how large language models can augment provider data workflows. Mentoring interns on complementary healthcare IT projects and supplementing rapid prototyping. Tech: Elixir, Phoenix, Python 3.x,
- **August 23, 2024 – December 2024:** Collaborated with interns on a Medicaid prediction research project, including foundational data science concepts and workflows. Led a follow-up imitative exploring cryptocurrency. Developing a Bitcoin indexer prototype. Project paused when students returned to academic commitments. Tech: macOS, Windows, EMRTS Linux network, GitHub, Python 3.x, Flask, Postgres.
  - *Interim contracts: CSpire, Jackson MS (Jul 2018–Jan 2019); Orchid Orthopedic Solutions, Remote (Feb 2020–Dec 2020); Quantuvos, Remote (Jul 2021–Feb 2022) — see entries below.*
- **July 2015 – January 2017:** Developed proof-of-concept tools and prototypes for processing medical transaction data (EDI_X12_837) and handling large medical provider datasets (JSON, XML). Extended a web services application for state governments and clinics to facilitate medical provider data registration and validation. **Tools:** Python 3.5, PostgreSQL, Django, JavaScript | **Platform:** Unix
- **October 2013 – March 2015:** Developed research-driven projects for mobile medical screenings and provider enrollment for state governments. Contributed to developing a Health Information Exchange(HIE) application for hospitals. **Tools:** Elixir/Phoenix | **Platform:** Ubuntu Linux, macOS | **Databases:** MariaDB, PostgreSQL, MongoDB
  - Interim contracts: Hilton, Memphis TN (Apr 2012–Aug 2012); Lowes Inc., Mooresville NC (Aug 2012–Jun 2013) — see entries below.
- **January 2012 – March 2012: Focused on electronic medical records, claims processing, and provider enrollment applications on GNU/Linux. Developed tools for handling JSON data and extended the application's support for additional records processing.**
  Tools: Python, Django, JavaScript (jQuery, Node.js) | Databases: MariaDB, PostgreSQL, Redis
  - *Interim contract: Hewlett-Packard, Nashville TN (May 2010–Dec 2011) — see entry below.*
- **July 2009 – May 2010:** Conducted code analysis for GNUmed, an open-source electronic medical record system, to evaluate its suitability for client needs. Built a RESTful web service for state medical provider validation. **Tools:** Python, Django | **Platform:** GNU/Linux

Quantuvos

Remote | Jul 2021 – Feb 2022

Technical consulting for Quantuvos, a coaching-platform startup. Reviewed and evaluated code for the client-facing MVP, supporting coaches and clients in scheduling workshops and sessions. Performed code reviews, identified issues, and recommended improvements. Contract position.

Orchid Orthopedic Solutions

Remote | Feb 2020 – Dec 2020 https://www.orchid-ortho.com

Contract work preparing Word, Excel, and PDF documents for archival submission to EtQ Reliance. Programmatically inserted tags and formatting in Word documents and validated spreadsheet metadata for accuracy and completeness. Contract position.

CSpire

Jackson, MS | July 2018 – Jan 2019 https://www.cspire.com

Led development of a RESTful auto-provisioning API for tracking and managing customer services delivered through CSpire and its third-party vendor partners. The application supported logging, reporting, and administration for both business office and network teams. Contract position.

Lowes Inc.

Mooresville, NC | Aug 2012 – Jun 2013 https://corporate.lowes.com

Refactored and extended legacy C code to support EMV (chip-based credit and debit card) transaction processing as part of Lowe's transition away from magnetic stripe. Work included code analysis, bug fixes, and documentation. Contract position.

Hilton

Memphis, TN | Apr 2012 – Aug 2012

Provided production support for Hilton's Central Reservation System. Contributed programming support using C, Perl, and shell scripting. **Tech:** Solaris, GNU/Linux, Informix, MySQL

Hewlett-Packard

Nashville, TN | May 2010 – Dec 2011

Performed software maintenance and troubleshooting for TennCare (State of Tennessee). Key projects included:

- Refactored and extended a SOAP server (written in C) for enhanced data exchange within the Medicaid Management Information System (MMIS).
- Developed monthly report generators for medical claims processing.
- Ported the entire application codebase to a new 64-bit Solaris network with updated compilers and Oracle database version.
- Contributed to technical writing and documentation.

Tech: C, PL/SQL, SOAP, Oracle, Solaris

ACS

Nashville, TN | Feb 2009 – Jun 2009

Migrated MMIS (Medicaid Management Information System) software for TennCare from Hewlett-Packard's infrastructure to ACS. Developed automation scripts and analyzed legacy C modules. Contract position.

Nationwide

Columbus, OH | Oct 2008 – Feb 2009 Analyzed annuity and financial data daily to detect anomalies in high-volume datasets. Proposed and developed a Python-based data validation and reporting tool to modernize and replace aging shell scripts used in production. **Tech:** Solaris, Oracle, proprietary scripting

Lexmark

Lexington, KY | May 2007 – Oct 2008 https://www.lexmark.com/en_us.html

Bug fixes, code analysis, and refactoring of C code for high-end prototype printers designed for secure environments (government, military, finance). Developed Python scripts for daily integration testing. Contract position.

IBM Internet Security Systems

Atlanta, GA | Jan 2007 – May 2007

Developed tools for porting and rebuilding C/C++ applications from Microsoft Visual Studio to Unix environments, using Visual Studio build artifacts rather than traditional makefile-based builds. Contract position.

Lowe's Inc.

Wilkesboro, NC | Feb 2006 – Dec 2006

Internationalized software for Lowe's first Canadian store. Maintained and extended existing C and shell scripting on AIX.

Tech: C, AIX, Oracle, DB2

Cingular Wireless

Alpharetta, GA | Aug 2004 – Feb 2006

Maintained and extended a legacy billing engine running on AIX and GNU/Linux using Oracle. The system was written primarily in C++ with shell scripts. Taught a C++ course to the code acceptance testing group and developed a Perl application that automated the configuration of their test boxes. **Tech:** C, C++, Perl, Oracle, AIX

SAIC

Oak Ridge, TN | Feb 2003 – Aug 2004 https://saic.com

Maintained and extended a legacy Perl application for network-wide resource and account management at Oak Ridge and Los Alamos National Laboratories until it was decommissioned.

University of Tennessee – Center for Environmental Biotechnology

Knoxville, TN | Aug 2002 – Dec 2002 https://ceb.utk.edu/ Built bioinformatics tools for monitoring, reporting, and analysis of microarray gene data. Tutored professional staff and graduate students in Perl programming for bioinformatics research.**Tech:** Perl, C, MySQL, Red Hat Linux

Manhattan Associates

Atlanta, GA | Jul 2001 – Jul 2002

Performed code maintenance on a cross-platform warehouse management system written in C++ for Linux and Windows. Contract position.

LHS/Sema Group Telecom

Atlanta, GA | Jan 2000 – Jul 2001

Performed code maintenance and enhancement for a large legacy billing engine written in C/C++ using Oracle, spanning multiple Unix platforms. Contributed to major technical documentation rewrites. **Tech:** C, C++, Oracle, Unix

Intellivoice / Premier Technologies

Atlanta, GA | Nov 1998 – Jan 2000

Worked on multimedia integration R&D for distributed Solaris networks. Developed coding projects using C++ and evaluated third-party middleware (such as CORBA) to facilitate network communication. **Tech:** C++, CORBA, Solaris

MCI/WorldCom

Atlanta, GA | Feb 1996 – Nov 1998

Full-cycle project to implement a billing engine on AIX using C++, Pro*C, and Oracle. Developed a Perl application that processed firearm purchase applications for the State of California (an MCI client). **Tech:** C++, SQL, Unix, Perl